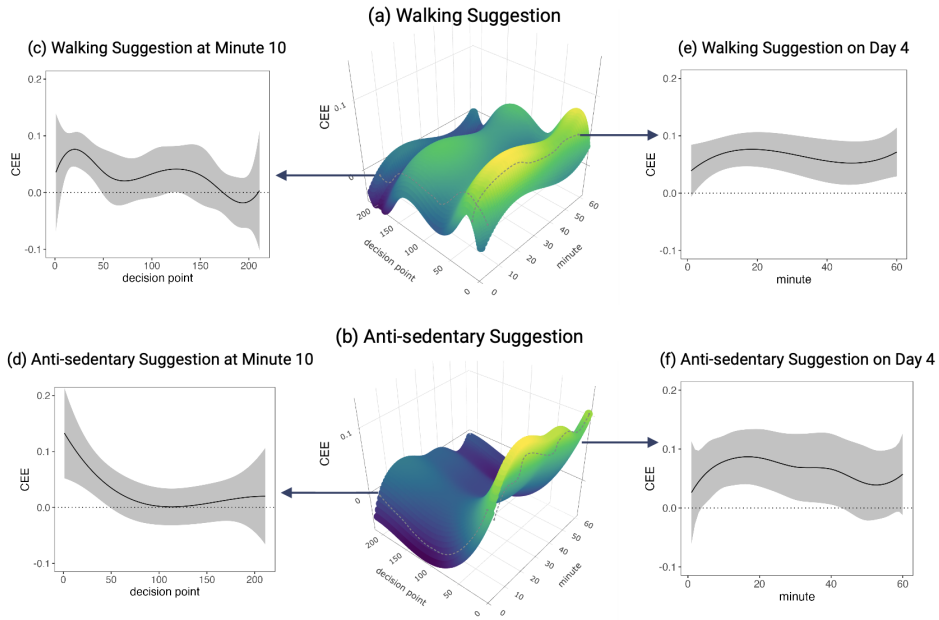

Figure 5: Surface plots for CEE of (a) walking suggestions and (b) anti-sedentary suggestions over decision points and minutes from the analysis in Section 7.1. A lighter color represents a larger effect. Panels (c) and (d) are slices of the surface plots at minute 10 after receiving the notification, illustrating the CEE over the course of the study period; panels (e) and (f) are slices of the surface plots at decision point 20 (day 4), illustrating the CEE over 60 minutes following the decision point.

within 30 minutes following a decision point. Figure 5(c) and (d) both show a decreasing CEE over decision points where the largest effect of both types of suggestions occurred early on in the study. In addition, we observed that compared with walking suggestions, anti-sedentary suggestions started with a greater effect but the effect declined much faster over decision points. When examining the effect over minutes at a fixed decision point (the plots showing day 4), significantly positive effects occurred around minute 20 for both types of suggestions, with the walking suggestion having a positive effect over a wider window up to minute 40 (Figure 5(e) and (f)). This observation aligns with the intervention design in that the suggested activities were brief, with the walking suggestions expected to take longer to complete than the anti-sedentary suggestions. The delay between receiving (minute 0) and acting upon the suggestion (the peak) is attributed to the *in situ* nature of the suggestions as they were integrated into the participants' daily lives.

7.2 Moderated Causal Excursion Effect

We considered the following CEE model for the effect moderated by a single time-varying covariate, S_{ij}:

$$\begin{aligned}&\mathbb{E}\{Y_{ijt}(\bar{A}_{i,j-1},k) - Y_{ijt}(\bar{A}_{i,j-1},0) \mid I_{ij}=1, S_{ij}\}\\ &= \beta_{01}^{(k)} + \beta_{02}^{(k)} S_{ij} + \beta_{11}^{(k)}(j) + S_{ij}\beta_{12}^{(k)}(j) + \beta_{21}^{(k)}(t) + S_{ij}\beta_{22}^{(k)}(t), \quad k=1,2. \qquad (10)\end{aligned}$$

We conducted three moderation analyses: (i) $S_{ij} = \text{loc}_{ij}$, which equals 1 if the participant was at home or work and 0 otherwise; (ii) $S_{ij} = \text{weekday}_{ij}$, which equals 1 if the j-th decision point was on a weekday and 0 otherwise; and (iii) $S_{ij} = \text{sedentary}_{ij}$, which equals 1 if the participant was sedentary (less than 56 steps in the 30 minutes before j-th decision point), using the median step count as the threshold.

Table 2: Selected number of knots for each functions in the moderator analysis.

Moderator	$\beta_{11}^{(1)}(j)$	$\beta_{12}^{(1)}(j)$	$\beta_{21}^{(1)}(t)$	$\beta_{22}^{(1)}(t)$	$\beta_{11}^{(2)}(j)$	$\beta_{12}^{(2)}(j)$	$\beta_{21}^{(2)}(t)$	$\beta_{22}^{(2)}(t)$
Weekday/Weekend	2	3	0	5	0	4	1	7
Location	3	1	0	7	0	7	3	4
Sedentary	1	4	2	4	1	6	3	6

The nuisance function $\eta(j, t, H_{ij})$ in Stage 1 and the functional β's in Stage 2 were fitted using the same approach as the marginal CEE analysis in Section 7.1, except that the corresponding moderator was included as an additional feature in Stage 1. The selected number of interior knots for the β's are in Table 2. To better interpret the results and answer the research questions raised at the end of Section 2, we calculated and presented the treatment effects conditional on various values of moderators.

Figure 6 shows the CEE profile over the 60 minutes following decision point $j = 35$ (day 7) from these moderation analyses. Figures for other decision points are included in Section A of the Supplementary Material, and they are vertical shifts of Figure 6 due to the additive nature of the model. Figure 6(a) shows that on weekdays, both the walking suggestion and the anti-sedentary suggestion had a significant positive effect around minutes 10-20, whereas on weekends neither suggestion had a positive effect. This indicates that structured environments may facilitate more routine responses; this could also be due to the suggestions being better tailored on weekdays. Figure 6(b) shows that only the walking suggestion exhibits a significantly positive effect in a narrow window around the peak of 10-20 minutes. Figure 6(c) shows that both suggestions were more effective when the participant was sedentary (compared to not) during the prior 20-40 minutes. The walking suggestion